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Land Use and Land Cover Prediction with QGIS-based Machine Learning Algorithms: A Case Study of Tha Chin River Basin, Thailand

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Abstract

Land use and land cover (LULC) changes have significant impacts on the environment, including water resources, biodiversity, and carbon storage. Accurate prediction of LULC changes is therefore essential for effective land management and environmental planning. The purpose of the study is to illustrate the importance of LULC alterations to the ecosystem. This study presents a case study of the Tha Chin River Basin in Thailand, using Random Forest classification from the Semi-Automatic Classification plugin to classify LULC classes and integrated cellular automata coupled with an Artificial Neural Network (CA-ANN) methodology within the Methods of Land Use Change Evaluation (MOLUSCE) plugin to predict LULC changes in QGIS. To produce land use and land cover (LULC) maps for different time periods, Landsat images from 2000, 2010 and 2020 were analyzed. The classification's performance was evaluated based on various criteria, including overall accuracy and kappa coefficient. The classified 2000 and 2010 maps were used to predict the year 2020. The predicted 2020 map was validated with the classified map. The results gave an accuracy assessment of over 80%. The LULC predictions showed a decline in forest cover and an increase in agriculture and urban areas. The study highlights the need of tracking and managing LULC changes in the Tha Chin River Basin and other analogous places while demonstrating the effectiveness of machine learning-based QGIS in properly predicting LULC.

Keywords: LULC prediction; Random Forest Classification; QGIS; SCP plugin; MOLUSCE plugin.

1. Introduction

Accurate land use and land cover forecasting is essential for various environmental management aspects. It provides critical information for disaster risk assessment, sustainable land management, and urban planning, enabling informed decision-making and effective policy implementation. Foley et al. (2005) highlight that land use forecasting supports sustainable land management by promoting effective land use planning and minimizing ecological footprints [1]. In urban planning, Chen and Yang (2010) stress the significance of land use forecasts in directing urban growth, maintaining efficient land use patterns, and mitigating issues like urban sprawl and inadequate infrastructure [2].

Furthermore, land use and land cover forecasting contribute to the management and conservation of natural resources. Lambin and Geist (2006) emphasize the importance of accurate projections in identifying vulnerable areas, implementing targeted conservation measures, and developing strategies for sustainable forestry, agriculture, and water resource management [3]. Additionally, land use forecasting plays a vital role in disaster risk assessment by helping stakeholders identify areas prone to natural calamities and enabling the adoption of suitable risk reduction strategies. Guo et al. (2019) emphasize its significance in disaster management frameworks [4].

Machine learning algorithms built on QGIS have proven to be quite useful for anticipating land use and land cover, offering insightful information about how the environment is changing over time. These algorithms offer a quick and accurate way to examine and forecast land use patterns, enabling well-informed land management and planning decisions.

The Semi-Automatic Classification Plugin (SCP) created by Congedo (2016) is a fantastic tool for QGIS [5]. Users can do supervised and unsupervised classification using SCP, which blends a variety of machine-learning algorithms and remote sensing approaches. Support vector machines (SVM) and random forests are two examples of the algorithms that SCP's customers can employ to categorize different types of land cover based on satellite imagery and other geospatial data. For predicting land usage and land cover, supervised classification using random forest is extremely useful. A machine learning system called a random forest mixes different decision trees to get precise predictions. It has been effectively applied in numerous mapping studies of land use and land cover. To forecast changes in land cover in urban areas, he employed random forests in QGIS [6]. The algorithm's ability to achieve high classification accuracy and capture intricate land cover dynamics was proved by this study.

The MOLUSCE plugin, developed by Lassalle et al. (2015) provides an advanced approach, along with SCP, for land use and land cover prediction [7]. It incorporates cellular automaton artificial neural networks (CA-ANN), offering a sophisticated method to accurately predict changes in land use and land cover patterns. By integrating land change models and machine learning techniques, MOLUSCE enables the simulation and prediction of land cover changes over time. This powerful tool combines land change models with the CA-ANN modeling technique, which leverages artificial intelligence to effectively capture the spatial dynamics and interactions among various land cover classes. This integration allows for accurate forecasting of changes in land cover patterns. MOLUSCE was used with CA-ANN and QGIS in a study by Chen et al. (2019) to forecast changes in land use in metropolitan areas [8]. This study showed how this method might anticipate future land use patterns and evaluate how various scenarios will affect changes to the land cover.

In the Tha Chin River Basin in central Thailand, there are a variety of landscapes that are prone to shifting land use and land cover. Understanding patterns and changes in land use and land cover within watersheds is crucial for sustainable resource management [9] since they are a major water resource system supporting many sectors like agriculture, industry, and domestic water supply. The basin region faces the issue of maintaining a balance between socioeconomic development and environmental conservation as urbanization, agricultural expansion, and infrastructural development continue to accelerate [10]. Planning for efficient land management and resource management requires accurate forecasting of changes in land use and land cover. By utilizing cutting-edge remote sensing methods and predictive modeling strategies, this conference document seeks to discuss research on forecasting land use and land cover in the Tha Chin River Basin. The study's projections help us gain a deeper understanding of how land use changes affect water supplies, biodiversity, and ecosystem services. The study will also assist stakeholders, including policymakers, land managers, and others, in creating efficient plans and guidelines for sustainable development in the Tha Chin Basin and comparable regions.

2. Materials and Methods

2.1. Study Area

The Tha Chin River Basin, which spans 13 provinces and 13,446.5 square kilometres, is one of Thailand's principal river basins. The Tha Chin River, a tributary of the Chao Phraya River, originates in Chai Nat Province and eventually drains into the Gulf of Thailand. The river is 0 to 25 meters high and 300 kilometres long. The Tha Chin River Basin has a population density of 2 million people. The region experiences annual rainfall averages of 1100 mm and water volumes of 1.35 billion cubic meters, respectively. The basin receives roughly 1,237 million cubic meters of water during the wet season, compared to about 120 million cubic meters during the dry season [11].

Figure 2 shows the exact location of the study area and outlines the Tha Chin River basin. The basin includes a substantial network of both natural and man-made channels, low flow, and a largely flat landscape. Intensive land usage, particularly for agriculture, industrialized animal husbandry, and the ongoing expansion of urban and industrial regions are characteristics of the region [12]. The Chao Phraya and Tha Chin River basins were significantly affected by the disastrous floods that struck Thailand in

2011. The Middle Tha Chin River Basin was chosen as the primary flood storage location as a result. This action was taken to lower flood peaks from the Chao Phraya and Tha Chin river basins and to store extra flood water. For migration and development to be facilitated in the Tha Chin River Basin, a full understanding of the river's hydrology and land use/cover is required. Due to the particular characteristics of the region and the rising needs of a growing population, efficient planning and management methods are made possible by this knowledge.

Fig. 1. Location Map of Study Area: Tha Chin River Basin

2.2. Data Collection

The data collection process for this study involved gathering various datasets from reliable sources. A digital elevation model (DEM) was obtained from the USGS Earth Explorer website, specifically the Shuttle Radar Topography Mission (SRTM) dataset, which provided a spatial resolution of 30 meters. This dataset was crucial for understanding the topographical characteristics of the study area. The road network data, acquired from the DIVA-GIS website, comprised the main road map of the Tha Chin Basin Area. This information was essential for assessing the transportation infrastructure and its potential influence on land use patterns. Satellite images from Landsat were acquired for the years 2000, 2010, and 2020, via the USGS Earth Explorer website. These images served as the primary source for generating land use/land cover maps. By utilizing these diverse datasets, encompassing elevation, transportation, and satellite imagery, a comprehensive foundation was established for conducting a robust analysis of land use and land cover changes in the Tha Chin River Basin.

Table 1. Data Collections and Data Sources

Data Type	Data Source	Data Description
Digital Elevation Model (DEM)	https://earthexplorer.usgs.gov	SRTM (Shuttle Rader Topography Mission with 30m spatial resolution)
Road Network	https://diva-gis.org	The main road map of Tha Chin Basin Area
Land Use/Land Cover Maps for 2000, 2010, and 2020	https://earthexplorer.usgs.gov	Satellite images from Landsat

2.3. Land Use/ Land Cover Classification

After extracting and projecting the research region, the land-use data subtypes were aggregated to establish five separate groups. Cities, farms, forests, water, and arid land. The following categories were listed: Canals, lakes, reservoirs, and rivers all contain water. Urban regions, rural communities, and other construction sites are all included in cities as built-up areas. Woods and shrubs made up the woodland. There are paddy fields, highland fields, farmed land, and arid regions devoid of any flora.

We employed the supervised classification technique, Random Forest of the SCP plugin for QGIS to accomplish land use classification. Using a collection of input variables, this method classifies different types of land cover using machine learning methods. Utilizing Landsat satellite imagery from the years 2000, 2010, and 2020, the classification, in this case, was carried out.

The resampling procedures in QGIS were used to assure consistency between the land use data and the spatial variables for modeling and predicting likely future transformation. These methods were used to correct any spatial resolution discrepancies between the spatial variables employed in the study and the land use data. The classification of land use types, including water, urban, forest, crop, and dryland, is made possible by this method, which also makes it easier to prepare data for modeling and subsequent forecasting.

2.4. Preparation of Spatial Variables

Three basic data types must be gathered and processed in order to prepare spatial variables: the slope, the digital elevation model (DEM), and the length of the route. The DEM dataset provided information on the elevation of the study area, allowing for the derivation of various topographic characteristics. Slope, a derivative from DEM, measures how steep the terrain is, giving important details about the ground's characteristics. The Euclidean distance approach was used to compute the distance to the road in order to account for the impact of the road network on the pattern of land use. By measuring the straight-line distance between each place in the study area and the closest road, this method allowed for an evaluation of the accessibility and proximity of various land regions to transportation infrastructure. These spatial variables were combined to create a complete dataset that captured significant topography and transportation aspects. This dataset would later be used as an input for transition potential modeling and prediction.

2.5. Evaluating Correlation

The evaluation of correlation involved in using the MOLUSCE plugin, a potent tool for modeling changes in land use and land cover. A correlation study was carried out to determine the relationship between different variables and changes in land use. In this approach, correlation values between patterns of land use change and a variety of explanatory variables, including socioeconomic information, environmental indicators, and geographic characteristics, were calculated. The evaluation of correlation is intended to identify important linkages and measure the strength and direction of these interactions. A rigorous statistical framework was used, and the proper significance tests were carried out, to guarantee the accuracy and dependability of the results. The evaluation of correlation utilized in the MOLUSCE plugin improved the scientific rigor and advanced our understanding of the subject of land-use/land-cover change analysis.

2.6. Area Change Analysis and Transition Potential Modelling

Analysis of area changes entails quantifying and characterizing changes in land use and land cover over a predetermined time period. By comparing land-use maps made at various times in time, MOLUSCE allows us to compute change metrics including gross change, net change, and transition matrices. These studies provide insight into the size, direction, and geographic distribution of land-use change, as well as tipping points and trends.

On the other hand, transition potential modelling focuses on predicting future land-use changes based on past trends and the influence of different factors. The MOLUSCE project integrates geographical and non-spatial datasets to determine the probability of a change in land-use category. By simulating the transition potential, which takes into consideration factors like proximity to metropolitan centers, environmental variables, and simple access to infrastructure, it is feasible to identify areas with a higher propensity for particular land-use changes.

2.7. Cellular Automata Simulation

The utilization of cellular automata is a crucial component of this module even if there are several ways to create a transition potential map. It makes use of cellular automata simulations, a type of computational intelligence in which artificial neural networks (ANNs) are an important component. LULC data that have been categorized serve as the basis for calibration and LULC modeling input data. This strategy is helpful when dealing with significant amounts of hazardous or challenging-to-implement input. A continuous index between 0 and 1 is created using ANN with fuzzy logic requirements to define the appropriateness of the terrain. The interactions and changes in weight connections between linked neurons make up the fundamental components of an ANN.

2.8. Validation

The MOLUSCE plugin relies on the kappa coefficient as a crucial validation indicator to determine how accurate the land use/land cover classification (LULC) is. This factor provides a quantitative indication of the degree of agreement between the reference LULC map or references truth and the predicted LULC map produced by the model. By comparing the predicted and reference LULC maps using a confusion matrix, the MOLUSCE plugin determined the kappa coefficient. The Confusion Matrix displays the pixels that were correctly and wrongly assigned to each land use classification. The value of the confusion matrix is then used to calculate the kappa coefficient. Kappa coefficient is represented by the following equation;

$$\boldsymbol{K = (P_o - P_e)/(1 - P_e)}$$

Where K represents the kappa coefficient, P_o is the observed proportionate agreement, and P_e is the proportionate agreement expected by chance. The observed proportionate agreement (P_o) is computed as the ratio of the total number of agreements between the predicted and reference classifications to the total number of observations. The proportionate agreement expected by chance (P_e) is calculated based on the marginal totals of the confusion matrix. It represents the expected probability of random agreement between the predicted and reference classifications. Both P_o and P_e values range from 0 to 1, with 1 indicating perfect agreement and 0 representing no agreement beyond what would be expected by chance alone. The kappa coefficient provides a standardized measure of agreement that takes into account chance agreement, making it a widely used metric for evaluating the accuracy of land-use classifications and other categorical assessments.

3. Results

3.1. LULC Change Analysis

The area of land use/ land cover change and the overall accuracy assessment for the supervised classified maps are described in Table-2.

Table 2. Area of Land Use/Land Cover Classes

Year	Area (km^2)					Overall Accuracy Assessment
	Water	Urban	Forest	Crop	Barren	
2000	1022.8	3241.7	4208.2	3185.5	1743.9	0.84
2010	1077.34	2955.2	2322.35	3882.97	3164.44	0.68
2020	947.41	3320	2960.43	2624.32	3550.12	0.71

a b c

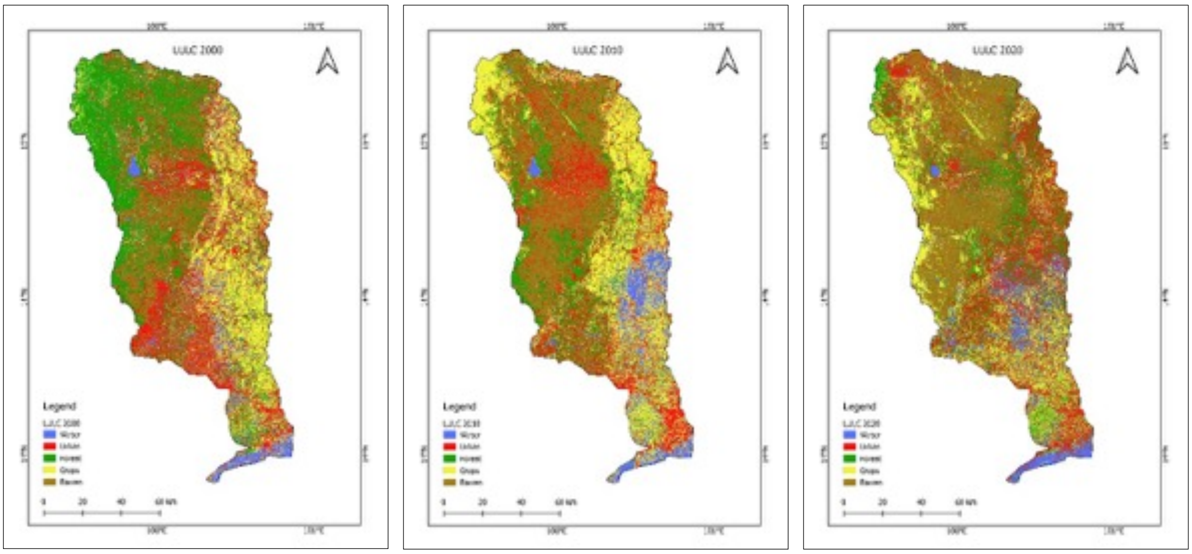

Fig. 2. LULC map for (a) the year 2000, (b) the year 2010, (c) the year 2020

According to Table 2, an analysis was conducted to assess the land use and land cover (LULC) changes in the study area over the years 2000, 2010, and 2020. The LULC maps for 2000 and 2010 were obtained from Landsat 5 satellite imagery, which exhibited limitations in terms of resolution and image distortions. Nevertheless, some key observations can be made. Urban areas showed significant expansion between 2000 and 2020, while the crop area decreased over the same period. The forest area also exhibited a decline over this time frame. In 2010, the urban area appeared notably large, and the crop area was more than that observed in 2000. The distortions in the aerial photo taken by the Landsat 5 satellite could have contributed to these discrepancies. The accuracy assessment for the 2010 classification was determined to be 0.68, suggesting some limitations in the classification results.

Table 3. Transition Matrix from 2000 to 2010

Year		2010				
	LULC Category	Water	Urban	Forest	Crop	Barren
2000	Water	0.95	0.01	0.01	0.03	0.01
	Urban	0.037	0.37	0.19	0.21	0.2
	Forest	0.03	0.19	0.26	0.25	0.27
	Crop	0.16	0.16	0.1	0.53	0.05
	Barren	0.01	0.17	0.11	0.05	0.66

Table-3 presents the transition potential matrix for the period from 2000 to 2010, indicating the probabilities of land use and land cover (LULC) category transitions. Among the categories, water and urban areas were found to be the most stable, with transition probabilities of 0.95 and 0.37, respectively. The forest category had a transition probability of 0.26, indicating a moderate level of stability. In contrast, the crop category showed a higher transition probability of 0.53, suggesting a significant change in land use. Barren land demonstrated the highest transition probability at 0.66, indicating its relatively dynamic nature. These transitions contributed to the conversion of land from water to urban areas (0.01) and from forest to crop areas (0.27). Furthermore, water and forest categories experienced a decrease, contributing 0.093 and 0.094, respectively, to the expansion of built-up land.

3.2. Projection and Validation

The MOLUSCE plugin integrated with Cellular Automata Artificial Neural Network (CA-ANN) approach was employed LULC data from 2000 to 2010 along with spatial variables to project LULC for 2020. A validation Kappa value showed 0.89, and after obtaining the projected LULC, the classified LULC map of 2020 and the projected 2020 were compared. The correctness percent showed 89.5% and an overall kappa value of 0.63. Table 4 and figure 3 show the classified and predicted maps and statistics for 2020.

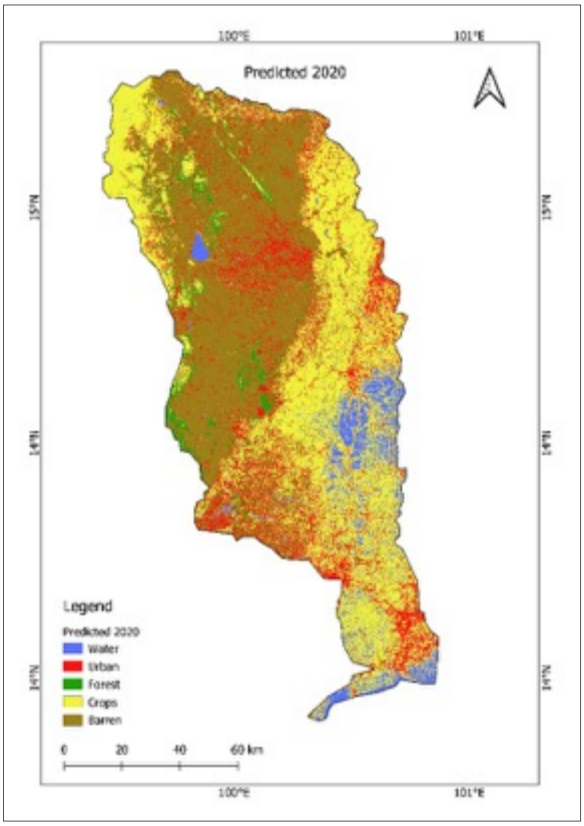

Fig. 3. Predicted LULC map for 2020

LULC Category	Classified LULC		Predicted LULC		Correctness	Kappa Value	
	Area(km²)	%	Area(km²)	%		ANN	Validation
Water	947.41	7.1	910.2	6.79	89.5	0.89	0.63
Urban	3320	24.8	3564.63	26.6			
Forest	2960.43	22.09	2453.7	18.31			
Crop	2624.32	19.58	2787.47	20.8			
Barren	3550.12	26.5	3684.27	27.5			

4. Discussion

An analysis of the Tha Chin Basin's land use and land cover change (LULC) from 2000 to 2020 finds numerous significant patterns. Over the same time span, the metropolitan area has grown significantly while the agricultural area has shrunk. The amount of forest is likewise declining. These findings illustrate how urbanization, altered agricultural methods, and potential deforestation have affected the study area.

It is important to be aware of the limits of using Landsat 5 satellite imagery, such as image resolution and distortion. The 2010 review of categorization accuracy revealed certain limitations in the results, probably brought on by distorted images. So, while interpreting the land cover classification for 2010, it is important to be careful. Understanding the LULC transition probabilities is possible by means of the transformation potential matrix. The water and urban regions had the greatest stability, whereas the crop type displayed a larger capacity for conversion, indicating major shifts in agricultural land use. The character of bare land is relatively changeable, either reflecting environmental degradation or ineffective land use techniques.

A high kappa validation value was achieved with the LULC 2020 projection utilizing the MOLUSCE plugin in conjunction with the CA-ANN approach. When comparing the projected and classified plots, considerable uncertainties were found, nevertheless, indicating that the anticipated LULC needs to be interpreted carefully. The correctness of the data, model presumptions, and outside influences are only a few examples of variables that can impact the predictability of results.

Overall, the findings emphasize the Tha Chin River Basin's potential for deforestation, fast urbanization, and alterations to agricultural land. The results highlight the significance of sustainable land management strategies, conservation initiatives, and additional studies to comprehend the causes and effects of these LULC shifts.

5. Conclusion

The Tha Chin Basin in 2020 was subjected to a thorough investigation of land cover change (LULC) in this study. Comparing the anticipated 2020 LULC map produced by the MOLUSCE plugin in QGIS with a classified LULC map taken from the same Landsat data allowed us to learn important details about the accuracy and dependability of its prediction model. The comparison of the structured and forecasted LULC maps revealed a high level of consistency, demonstrating the efficacy of the strategy used in this investigation. In order to produce accurate predictions, the random forest algorithm used in the SCP plugin proved to be resilient in identifying intricate correlations between spectral data and land cover layers. Validation points provided conclusive data to assess the correctness and precision of the proposed LULC map. This work highlights the potential of utilizing advanced classification methods in conjunction with Landsat imagery to track and comprehend changes in the LULC. A potent framework for evaluating land use patterns and enabling evidence-based decision-making for sustainability and natural resource management is provided by the integration of remote sensing, GIS technologies, and machine learning algorithms. Future research should focus on overcoming the issues raised by this work, such as enhancing the classification algorithm, expanding the quantity and quality of training data, and estimating the likelihood of inaccuracy in image interpretation. Additionally, carrying out more validation tests and incorporating real-world data can raise the predictive model's accuracy and dependability for use in the future. In conclusion, the comparison of the 2020 LULC maps that have been planned and those that have been categorized shows how well Landsat imagery and sophisticated classification methods can be used to evaluate changes in land cover in the Tha Chin Basin. Despite some discrepancies, the findings offer insightful information on the region's changing landscape and highlight the necessity of sustainable land management techniques to protect ecological integrity and the socioeconomic well-being of the area. In the Tha Chin River Basin and other similar areas around the world, we can encourage educated decision-making and undertake tailored interventions that balance human needs with environmental protection.

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